SUMING LI

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Computational design and robotic fabrication, with experience in structural form-finding, deployable structures, and robotic weaving. Research interests include origami, folding, and deployable architectural systems.

EDUCATION

Tongji University, Shanghai, China
Bachelor of Architecture, July 2026 | GPA: 4.31/5.0

Selected Coursework:
AI-driven Spatial Intelligence-Based Tectonics
Open Source Hardware and Programming
Computer Graphics

PUBLICATIONS & MANUSCRIPTS

Design and Implementation Methods for Multi-Stable Reconfigurable Rigid-Panel Origami Structures Based on Geometric Derivation and Physical Simulation
Ueki Wong, **Suming Li**, Yunning Feng, Junwang Yu, Yige Liu, and Philip F. Yuan. 2025.
Architectural Society of China Computational Design Academic Conference (CDAC 2025). Accepted conference paper (in Chinese); publication pending.

From Fold to Full-Scale: A Computational Design-Analysis-Fabrication Framework for Rigid Thick-Panel Deployable Structures
Yunning Feng, Junwang Yu, **Suming Li**, Linxu Xie, Ueki Wong, and Philip F. Yuan. 2026.
IASS Annual Symposium and 3rd Italian Workshop on Shell and Spatial Structures (IASS–IWSS 2026). Accepted conference paper; publication pending.

Design, Simulation, and Construction Path for Multi-Stable Reconfigurable Thick-Panel Deployable Structures
Yunning Feng, Junwang Yu, Ueki Wong, **Suming Li**, and Philip F. Yuan. 2026.
Humanistic Computation & Intelligence: Proceedings of the 31st International Conference on Computer-Aided Architectural Design Research in Asia (CAADRIA 2026), Vol. 2, pp. 553–562. DOI: 10.52842/conf.caadria.2026.2.553.

A Review of User Experience Evaluation Methods for Intelligent Virtual Agents: A Study of Multi-dimensional Frameworks and Application Scenarios
Ueki Wong, **Suming Li**, and Xueyang Wang. 2025.
Proceedings of the 13th International Conference on Human-Agent Interaction (HAI 2025), pp. 456–457. Conference poster paper. DOI: 10.1145/3765766.3765862.

Investigating Questioning Strategies for Robot Learning by Asking in Household Tasks
Yuanda Hu, Dai Shi, Jiani Hou, **Suming Li**, and Yujie Qiu.
Completed manuscript; not yet submitted.

RESEARCH EXPERIENCE

College of Architecture and Urban Planning (CAUP), Tongji University | Shanghai, China
| Jul 2026 to present

Research Assistant
Computational design research on multi-objective optimization for robotic weaving and semantic Gaussian splatting for architectural representation.

Selected Projects
Wicker Weaving Path Optimization
Developed a multi-objective path-design workflow for robotic wicker weaving, integrating topology optimization and stress-guided path generation.
Used Pareto-based search to balance material distribution, stress alignment, bending cost, and rod count under fabrication constraints.
Semantic Gaussian Splatting
Explored semantic Gaussian splatting for scene understanding and its potential use in robotic localization.

Digital Design and Research Center (DDRC), Tongji University | Shanghai, China
| November 2024 to June 2025

Research Intern
Supported research in robotic construction and human–machine collaboration, with responsibilities spanning system integration, debugging, validation, and research communication.

Selected Project
Robotic System Development & Integration
Assembled and debugged robotic arm end-effectors and construction-related hardware
Participated in system configuration, calibration, and functional validation

Research Communication & Publication Support
Prepared technical presentations for research progress reviews
Contributed to academic writing by documenting methodologies, conducting literature reviews, and assisting with experimental data analysis

Center for Digital Innovation, Tongji University | Shanghai, China | Jul 2025 to Oct 2025

Research Intern
Conducted robotics and computational design research with a focus on robot learning, human–robot interaction, and experimental system development. Contributed to both academic publication preparation and physical robotic prototyping.

Selected Projects
Investigating Questioning Strategies for Robot Learning by Asking in Household Tasks

Research & Academic Writing
Authored the Related Work section through a systematic literature review on robot learning and interactive perception
Produced publication-quality diagrams and figures for academic dissemination
Assisted in experimental framework design and evaluation planning

Robotic System Development
Participated in robotic platform construction, including mechanical assembly and sensor integration
Conducted functional testing and supported experimental data collection

TEACHING EXPERIENCE

DigitalFUTURES 2026 | Shanghai, China | Jun 2026

Embodied Weaving: Intelligent Fabrication of Natural Materials

Teaching Assistant
Developed the workshop weaving-design workflow, linking path generation, fabrication constraints, and indexed LED targets.
Collected and organized multi-view demonstrations for four robotic weaving primitives in a VLA-based system: picking, upper winding, lower winding, and pressing.

ACADEMIC CONFERENCES & WORKSHOPS

Computational Design Symposium & Annual Conference of the Computational Design Academic Committee of the Chinese Architectural Society | Shanghai, China | Nov 2024

Multi-Robot Collaborative Autonomous Construction of Reciprocal Shells

Outstanding Participant Award
Developed parametric reciprocal-frame geometries in Grasshopper/Kangaroo and detailed slotted components for physical assembly.
Translated manual interlocking trials into three repeatable actions (insertion, lifting, and bypass) and collected multi-view demonstrations for robotic task learning.
Co-built a complete reciprocal-shell prototype and contributed component details to a three-robot assembly demonstration.

DigitalFUTURES 2025 | Shanghai, China | Jun 2025

Exploring Deployable Structures for Architectural Design with Vector-based Graphic Statics
Developed parametric crease patterns and Kangaroo folding simulations for an umbrella-type deployable structure; evaluated load paths across folding states using vector-based graphic statics (VGS).
Detailed hinges, panel clearances, and cable-assisted deployment; co-built a working prototype using aluminum frames, polycarbonate panels, and steel hinges.

ADDITIONAL PROJECT EXPERIENCE

Tongji University Students Innovation Training Program (SITP) – Shanghai, China | 2022

AI-Powered Chinese Character Visualization System
Assisted in generative logic design and system-level planning

Intelligent Drainage Pipeline Defect Detection System
Contributed to mechanical form design of inspection hardware

AWARDS

First Prize, Intelligent Cultural Creativity & Innovation Competition
26th China Robotics and AI Competition (National Finals) | Aug 2024

Third Prize, China College Computer Competition: Intelligent Interaction & Innovation
National Finals | Oct 2023

Second Prize, RAICOM Robotics Developer Competition
National Finals | Aug 2023

Third Prize, Undergraduate Outstanding Student Scholarship
Tongji University | Dec 2025

PROFESSIONAL EXPERIENCE

JLL (Jones Lang LaSalle), JLL Technologies (Technology Business Line) | Shanghai, China | Mar 2026 to May 2026

Digitalization Solution Intern
Produced digital-solution presentations and client proposal materials; researched AR navigation and smart-locker products, communicated with vendors, and consolidated client requirements

BILIBILI, Vsinger Business Unit, Shanghai, China | Aug 2024 to Nov 2024

Project Management Intern
Contributed to design development and installation of the Luo Tianyi immersive exhibition; coordinated venue arrangements and organizational logistics for the resonance concert

TECHNICAL & SOFTWARE SKILLS

Programming Languages: Python, C++
Design & Modeling: Rhinoceros, Grasshopper, Kangaroo, AutoCAD
Graphic Design: Adobe Photoshop, Adobe Illustrator, Adobe InDesign

LEADERSHIP & SERVICE

Student Journalism Team, Publicity Department, Tongji University | Sept 2021 – Sept 2022

Content Creator
Planned and produced video content for university-wide communications and outreach initiatives.

Student Union, Tongji University | Sept 2022 – Jan 2023

Student Affairs Assistant
Supported administrative operations, meeting coordination, and campus event organization.

Student Union Societies, Tongji University | Sept 2022 – Sept 2023

Activities Manager
Coordinated student organization activities and led visual design initiatives for campus programs